

TNA Report – ATRIUM Computer Vision in Archaeology Training School
Institute of Archaeology of the Czech Academy of Sciences, Brno (ARÚB), 14–18 September

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Between 14 and 18 September 2026, I attended the advanced Training Course on 'Computer Vision in Archaeology', organised by the Archaeological Information System of the Czech Republic (AIS CR) at the Institute of Archaeology of the Czech Academy of Sciences in Brno. The event was organised jointly as part of the EU-funded 'Advancing Frontier Research in the Arts and Humanities' (ATRIUM) project and the COST Action 'Managing Artificial Intelligence in Archaeology' (MAIA) [CA23141].

This five-day intensive training provided a brief introduction to the subject of computer vision in archaeology and its applications for the detection, classification and analysis of archaeological material. The different presentations within the study blocks (microscopy, remote sensing, rock art and numismatic studies) served to explain the research possibilities, demonstrate the methods and provide practical training for participants, thanks also to prepared datasets.

The first day served as an introduction to the subject matter, as well as an opportunity to get to know the trainers and the other participants. Petr Pajdla delivered an introductory session on the fundamental concepts of Computer Vision and its applications in archaeology. During this presentation, emphasis was placed on the workflow, possibilities and limitations of computer vision in archaeology. This first session also served to encourage self-reflection, participatory work and discussions amongst the different groups of participants. Ronald Harasim presented agent-based programming to help participants understand how to programme and what to avoid when working with agents, whilst Filip Hájek gave an introduction to the fundamentals of Python. The second day of training focused on artefact image datasets and image annotation. Data preparation and annotation were explained and put into practice using CVAT. ArchaeoTag was also used as a more dynamic activity to reinforce knowledge related to annotation. The day concluded with presentations of various projects by some of the participants, followed by a guided tour of Brno's historic centre.

The third training session, led by Anastasia Eleftheriadou, focused on microscopic images, including image analysis, data set preparation, standardisation and automated segmentation. The dataset used during the practical session to consolidate the knowledge gained comprised a dataset derived from the analysis of wear patterns on stone tools and thin sections of sediment. On Wednesday, participants

were also treated to a guided tour of the laboratories of the Institute.

Thursday began with a training session led by Filip Hájek on satellite imagery. The practical session covered the application of automated techniques for the detection of archaeological features on land surfaces. In the afternoon, the focus shifted to rock art datasets, led by Ashley Green. We used Label Studio to create annotations, as well as a pre-trained YOLO segmentation model to detect cupmarks in archaeological images. On the final day, Ronald Harasim demonstrated how to apply computer vision workflows to datasets on coins. LLMs and vision-language models were also presented.

This training was a very valuable experience, offering a concise yet thorough introduction and an overview of the fundamentals of computer vision in archaeology. It provided an opportunity to reflect on the various potential uses, taking into account both the possibilities and the limitations of such applications.

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